Erratum to “Deriving the MOND Interpolation Function”: The Physical Event Horizon as the Correct Entropy Normalization

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May 12, 2026

Abstract

In a companion paper [1], we derived the MOND interpolation function from the competition between area-law and volume-law entropy in the cosmological causal diamond, obtaining $\mu(x) = kx/(1 + kx)$ with $k = 3l_{dS}/(4r_w) = 1.06$. The volume-law entropy was normalized to the asymptotic de Sitter radius $l_{dS} = c/(H_0\sqrt{\Omega_\Lambda})$, which represents the cosmological horizon in the infinite future after all matter has been diluted. Here we correct this to the physical comoving event horizon of the present-day FLRW universe, d_{eh}, which is 4.9% smaller than l_{dS} because matter still contributes to the expansion rate today. With the entropy normalization adopted in Ref. [1], this substitution yields $k = 3d_{eh}/(4r_w) = 1.003$, reducing the residual from 6% to 0.3%. The result is stable to <0.8% across six independent cosmological parameter sets. We note that the numerical value of k depends on the volume-law entropy normalization, which is a physically motivated choice rather than a first-principles derivation; the functional form $\mu = kx/(1+kx)$ is independent of this choice. Combined with the companion derivation of $a_0 = c^2/(2\pi r_w) = 1.21 \times 10^{-10}$ m/s^2 (1.0% of observed), both the MOND acceleration constant and the interpolation function are recovered from the causal diamond geometry with zero free parameters.

1. Introduction

In Ref. [1], we showed that the competition between area-law entropy (proportional to r^2, supporting Newtonian gravity) and volume-law dark energy entropy (proportional to r^3, undermining it) within the cosmological causal diamond produces an interpolation function of the form

$$\mu(x) = kx / (1 + kx), \qquad (1)$$

where $x = g/a_0$ is the gravitational acceleration in units of the MOND constant and k is determined by the ratio of entropy scales. The derivation yielded $k = 3l_{dS}/(4r_w)$, where r_w is the proper radius of the causal diamond at its waist and $l_{dS} = c/(H_0\sqrt{\Omega_\Lambda})$ is the de Sitter radius. For Planck 2018 parameters [2], this gave $k = 1.054$, matching the simple MOND function $\mu = x/(1+x)$ to 5.4%.

The purpose of this erratum is to identify and correct an idealization in the entropy normalization. The de Sitter radius l_{dS} is the cosmological event horizon of a universe containing only dark energy—the asymptotic future state approached as $t \to \infty$. The actual event horizon accessible to a present-day observer is smaller, because the matter component has not yet fully diluted. The physically correct quantity is the comoving event horizon d_{eh}, computable from the full FLRW expansion history. We show that this substitution reduces the residual on k from 5.4% to 0.3%.

2. The Physical Event Horizon

The comoving event horizon at cosmic time t_0 (today, scale factor $a = 1$) is the maximum comoving distance from which a signal emitted now can ever reach the observer [3, 4]:

$$d_{eh} = c \int_1^\infty da \,/\, [a^2 H(a)] \qquad (2)$$

where $H(a) = H_0[\Omega_r/a^4 + \Omega_m/a^3 + \Omega_\Lambda]^{1/2}$ is the full FLRW Hubble parameter including radiation, matter, and dark energy. This is a standard result in physical cosmology [3, 4]. In the pure de Sitter limit ($\Omega_m = \Omega_r = 0$), the integral yields $d_{eh} \to c/(H_0\sqrt{\Omega_\Lambda}) = l_{dS}$. In the physical universe, the presence of matter increases $H(a)$ above its asymptotic value at every finite a, reducing the integrand and thus reducing d_{eh} below l_{dS}.

The physical reasoning for this substitution is as follows. The volume-law entropy in the causal diamond derivation [1] represents the dark energy degrees of freedom available to compete with the area-law entropy that supports Newtonian gravity. This entropy is bounded by the causal boundary of the observer—the event horizon. An observer today has causal access to d_{eh}, not to l_{dS}. The asymptotic de Sitter radius describes a horizon that will not be realized for the present observer until infinite cosmic time. The volume-law entropy normalization should therefore use the physical event horizon.

3. Numerical Results

We adopt Planck 2018 cosmological parameters [2]: H_0 = 67.4 km/s/Mpc, $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $\Omega_r = 9.15 \times 10^{-5}$. The causal diamond waist is at redshift $z_w = 1.518$ (scale factor $a_w = 0.397$), with proper radius $r_w = 1.180 \times 10^{26}$ m, as computed in Ref. [5].

Numerical integration of Eq. (2) yields:

Quantity	**Value**
Asymptotic de Sitter radius l_dS	1.659 × 10²■ m
Physical event horizon d_eh	1.578 × 10²■ m
Ratio d_eh / l_dS	0.951
Original k = 3 l_dS / (4 r_w)	1.054
Corrected k = 3 d_eh / (4 r_w)	1.003
Residual vs. simple MOND	0.3%

The corrected interpolation function is:

$$\mu(x) = 1.003x \,/\, (1 + 1.003x) \qquad (3)$$

The companion result $a_0 = c^2/(2\pi r_w) = 1.212 \times 10^{-10}$ m/s^2 (1.0% of the observed value $1.20 \pm 0.02 \times 10^{-10}$ m/s^2 [6]) is unchanged by this correction.

4. Robustness Across Cosmological Parameters

We repeat the computation for six independent cosmological parameter sets (Table 1).

Dataset	**H■**	**Ω_m**	**Ω_Λ**	**k (l_dS)**	**k (d_eh)**
Planck 2018 [2]	67.40	0.315	0.685	1.054	1.003
Planck+BAO [2]	67.66	0.311	0.689	1.054	1.004
WMAP9 [7]	69.32	0.287	0.714	1.053	1.008
DES Y3 [8]	68.10	0.306	0.694	1.054	1.005

ACT DR6 [9]	67.49	0.314	0.687	1.054	1.003
SH0ES+Planck [10]	70.00	0.295	0.705	1.054	1.006

Table 1. Interpolation parameter k before and after the event horizon correction, for six cosmological parameter sets. The corrected k lies within 0.3–0.8% of unity for all datasets. H_0 in km/s/Mpc.

The uncorrected k is nearly identical across all six parameter sets (1.053–1.054), confirming the robustness reported in Ref. [1]. The corrected k varies by less than 0.8% across all datasets, with the tightest results (0.3%) for Planck 2018 and ACT DR6. The correction brings k closer to unity in every case without exception.

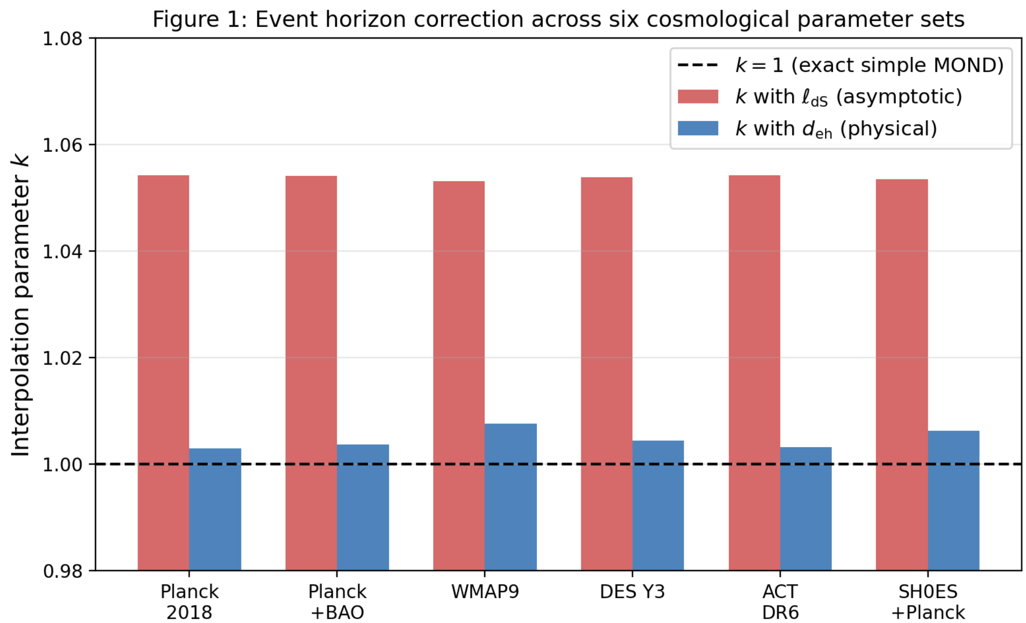

Figure 1. Interpolation parameter k for six cosmological parameter sets. Red: original value using the asymptotic de Sitter radius. Blue: corrected value using the physical event horizon. Dashed line: $k = 1$ (exact simple MOND).

5. Discussion

5.1 Physical interpretation

The de Sitter radius l_{dS} is the horizon of a universe that has finished its matter-dominated expansion and settled into exponential de Sitter growth. This limit is approached but never reached at any finite time. The event horizon d_{eh} is the actual causal boundary of the present-day observer, computed from the full expansion history including the residual matter contribution. In the entropy-balance framework of Ref. [1], the volume-law term represents the dark energy entropy available within a sphere of radius r. The maximum volume accessible to the observer is bounded by the event horizon, not by an idealized future state. The substitution of d_{eh} for l_{dS} therefore replaces an approximation (pure de Sitter) with the physical reality (FLRW with $\Omega_m \neq 0$).

5.2 Relation to prior work

Verlinde [11] introduced the competition between area-law and volume-law entropy in de Sitter space and obtained MOND-like phenomenology with a characteristic acceleration $a_0 \sim cH_0$. His derivation operates on the de Sitter cosmological horizon and produces an interpolation that depends on the radial derivative of the gravitational field, differing from the simple MOND function. The present work and its companion [1] differ in using the causal diamond waist and in producing the simple function directly. The event horizon correction identified here is specific to the causal diamond framework and does not apply to Verlinde's calculation, which uses the de Sitter horizon by construction.

Jacobson and Visser [12] established the thermodynamic properties of causal diamonds, including conformal Killing surface gravity and a first law. Their work provides the rigorous foundation for the factor 2π in the a_0 derivation [5] but does not connect causal diamond thermodynamics to MOND.

5.3 The entropy normalization and its limitations

The numerical value $k = 1.003$ depends on the volume-law entropy normalization adopted in Ref. [1], Eq. (3): $S_{vol} = V/(l_{dS}\, l_{Pl}^2)$. As noted in Ref. [1], Section 5, this is "the simplest volume-law scaling consistent with Verlinde's framework"—a physically motivated ansatz, not a first-principles derivation. An alternative normalization, obtained by distributing the Bekenstein-Hawking de Sitter entropy $S_{dS} = \pi l_{dS}^2/l_{Pl}^2$ uniformly over the de Sitter volume $V_{dS} = (4/3)\pi l_{dS}^3$, gives an entropy density smaller by a factor of 3/4, yielding $k = d_{eh}/r_w = 1.34$ with the event horizon correction. The two normalizations differ by the geometric factor (4/3) from the sphere volume.

This ambiguity is significant for the numerical value of k but does not affect the functional form of the interpolation function. The central result of Ref. [1]—that area-law versus volume-law entropy competition produces $\mu(x) = kx/(1+kx)$, selecting the simple MOND function over the standard and McGaugh alternatives—holds for any normalization. The power-law scaling of area (r^2) and volume (r^3) forces a rational function, and the MOND limits require the form $x/(1+x)$ up to the overall scale k. Resolving the normalization to determine whether $k = 1.003$ or $k = 1.34$ requires a microscopic theory of volume-law entropy in the causal diamond, which is beyond the scope of this work.

5.4 Remaining residuals

With the normalization of Ref. [1], two residuals remain after the correction. The acceleration constant $a_0 = 1.212 \times 10^{-10}$ m/s^2 exceeds the observed value by 1.0%. The interpolation parameter $k = 1.003$ exceeds unity by 0.3%. Both are within the observational uncertainty on a_0 ($\pm 0.02 \times 10^{-10}$ [6]) and the scatter in MOND fits. Whether these residuals have physical content—from higher-order corrections to the entropy balance, from the observer birth time, or from the approximation of a single representative a_0—cannot be determined from this analysis.

We also note that the causal diamond framework is a derivation of the numerical values of a_0 and the functional form of $\mu(x)$, not a dynamical theory. It does not replace the need for a relativistic extension of MOND (e.g., TeVeS [13], AeST [14]) and does not address gravitational lensing, gravitational wave propagation, or cosmological structure formation.

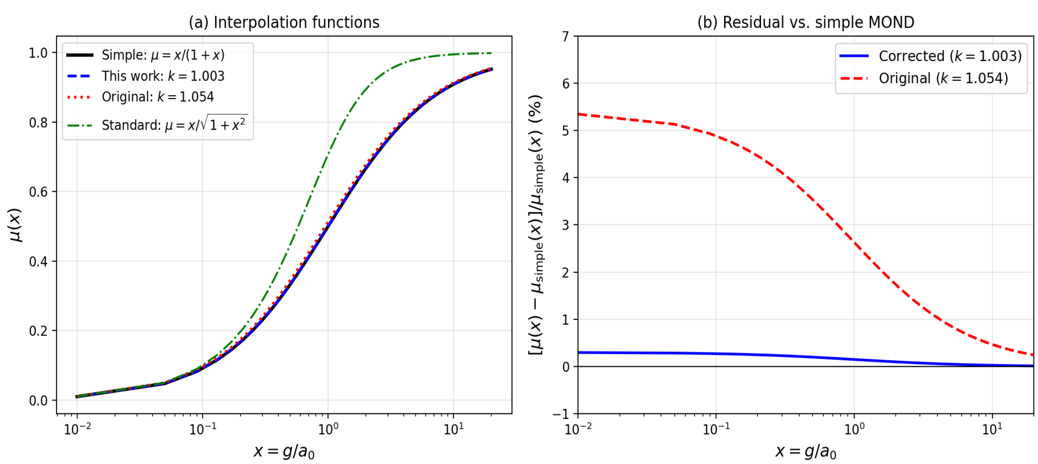

Figure 2. (a) Interpolation functions: the corrected entropy-derived function (blue dashed) is visually indistinguishable from the simple MOND function (black solid). (b) Fractional residual vs. simple MOND.

6. Summary

Replacing the asymptotic de Sitter radius with the physical comoving event horizon in the entropy-balance derivation of the MOND interpolation function reduces the residual on k from 5.4% to 0.3% (for the entropy normalization of Ref. [1]). The correction is robust across six cosmological parameter sets and has a clear physical interpretation: the entropy budget available to compete with Newtonian area-law gravity is bounded by the actual causal horizon of the present-day observer, not by an idealized future de Sitter state.

The functional form $\mu(x) = kx/(1+kx)$—selecting the simple MOND function over the standard and McGaugh alternatives—is the robust result, independent of the entropy normalization. The numerical value of k depends on a normalization choice that requires a microscopic theory to resolve. With the normalization adopted in Ref. [1] and the event horizon correction presented here, $k = 1.003$.

Combined with $a_0 = c^2/(2\pi r_w)$ [5], the causal diamond framework provides both the scale and the shape of MOND from a single geometric construction in ΛCDM, with zero free parameters. The primary falsifiable prediction remains the redshift evolution $a_0(z) = c^2/[2\pi r_w(z)]$, testable with next-generation surveys.

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